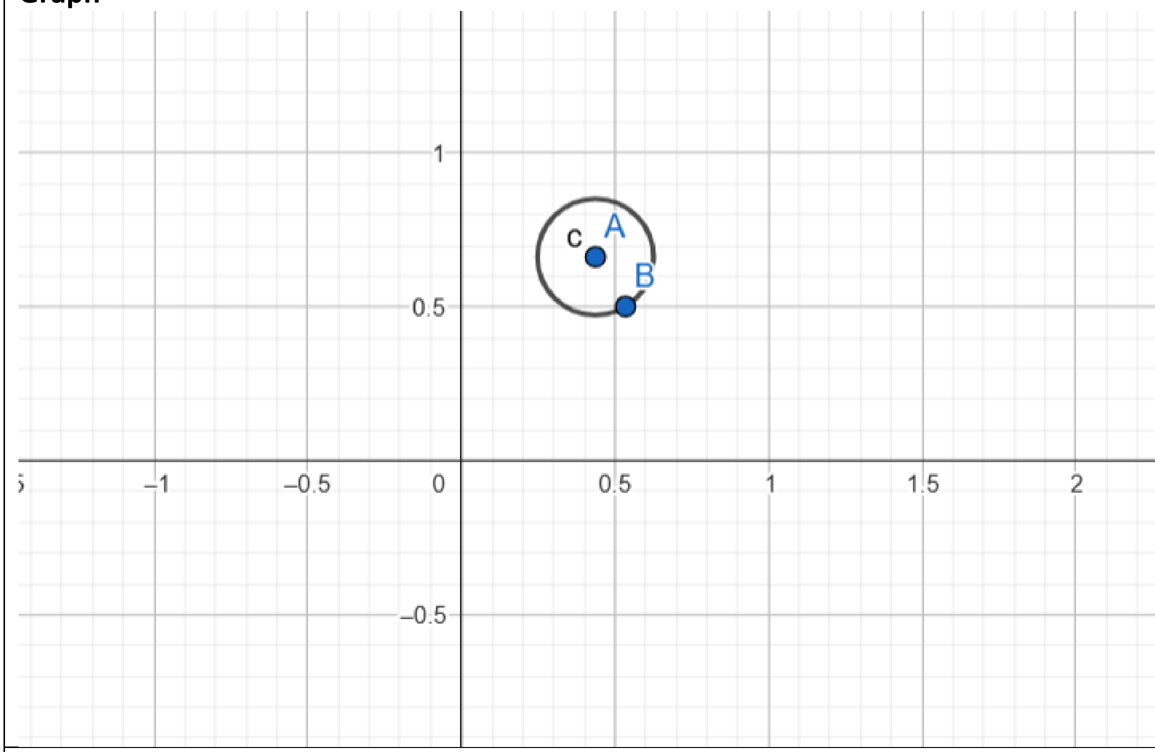
Lab Practice-7

Submission Guidelines-

- Rename the file with your serial number only
- Must submit within time that will be discussed in class VUES
- Must include resources for all the section in the table

Question-
Create a simple day and night scenario that will automatically change from day to night

Graph



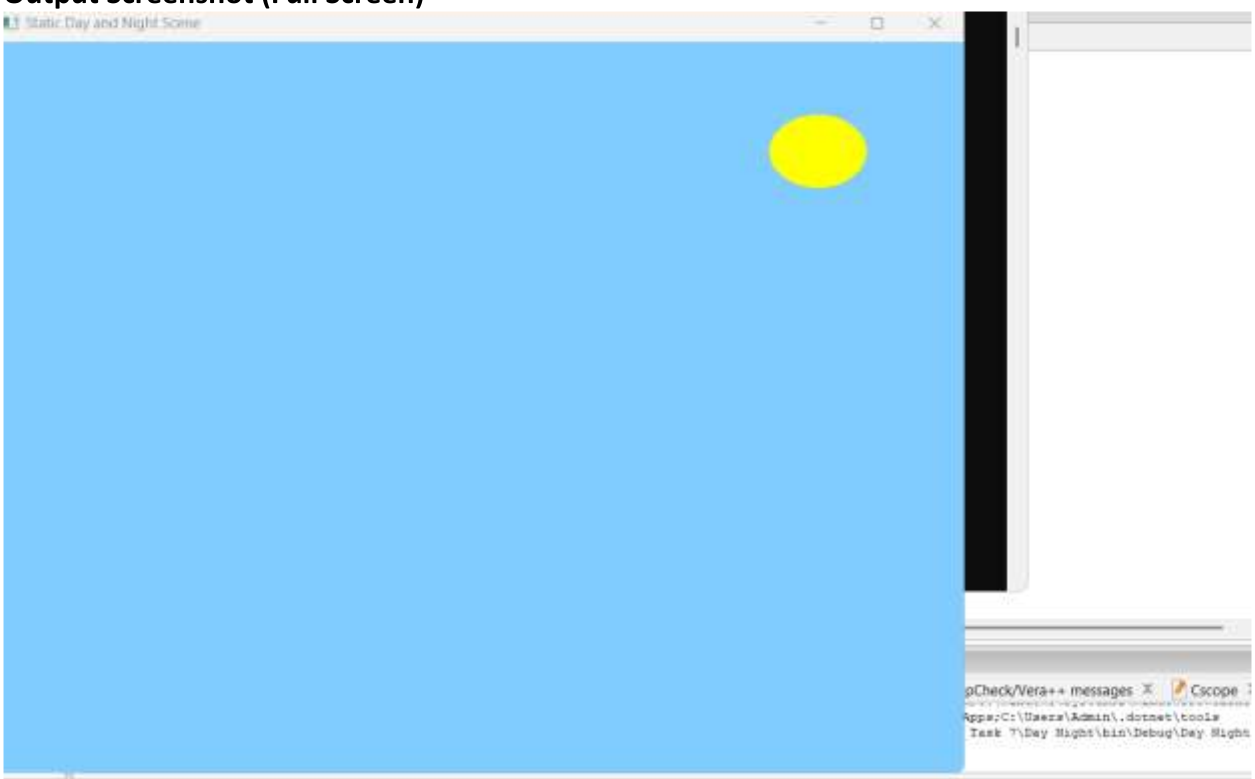
Code-

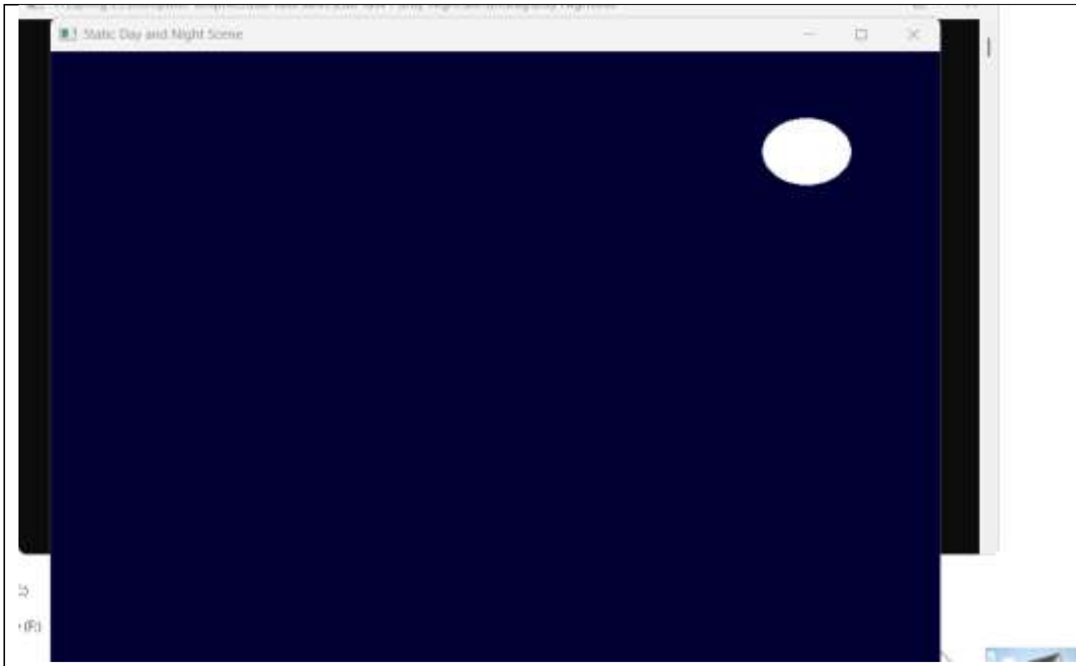
```
#include <GL/glut.h>
#include <cmath>
using namespace std;

bool isDay = true; // Start with day

void drawCircle(float x, float y, float radius) {
   int triangleAmount = 50;
   float twicePi = 2.0f * 3.1416f;
```

```
    glBegin(GL_TRIANGLE_FAN);
    for(int i = 0; i <= triangleAmount; ++i) {
        glVertex2f(
            x + (radius * cos(i * twicePi / triangleAmount)),
            y + (radius * sin(i * twicePi / triangleAmount))
        );
    }
    glEnd();
}

void display() {
    glClear(GL_COLOR_BUFFER_BIT);

    // Set background color
    if (isDay) {
        glClearColor(0.5f, 0.8f, 1.0f, 1.0f); // Day: sky blue
    } else {
        glClearColor(0.0f, 0.0f, 0.2f, 1.0f); // Night: dark blue
    }
    glClear(GL_COLOR_BUFFER_BIT);

    // Draw sun or moon
    if (isDay) {
        glColor3f(1.0f, 1.0f, 0.0f); // Yellow sun
    } else {
        glColor3f(1.0f, 1.0f, 1.0f); // White moon
    }

    drawCircle(0.7f, 0.7f, 0.1f); // Static position in top-right

    glutSwapBuffers();
}

void timer(int value) {
    isDay = !isDay; // Toggle day/night
    glutPostRedisplay(); // Redraw screen
    glutTimerFunc(3000, timer, 0); // Call again after 3 seconds
}

void init() {
    glMatrixMode(GL_PROJECTION);
    gluOrtho2D(-1, 1, -1, 1); // 2D coordinate system
}
```

```
int main(int argc, char** argv) {
    glutInit(&argc, argv);
    glutInitDisplayMode(GLUT_DOUBLE | GLUT_RGB);
    glutInitWindowSize(800, 600);
    glutCreateWindow("Static Day and Night Scene");

    init();
    glutDisplayFunc(display);
    glutTimerFunc(0, timer, 0);
    glutMainLoop();

    return 0;
}
```

Output Screenshot (Full Screen)-

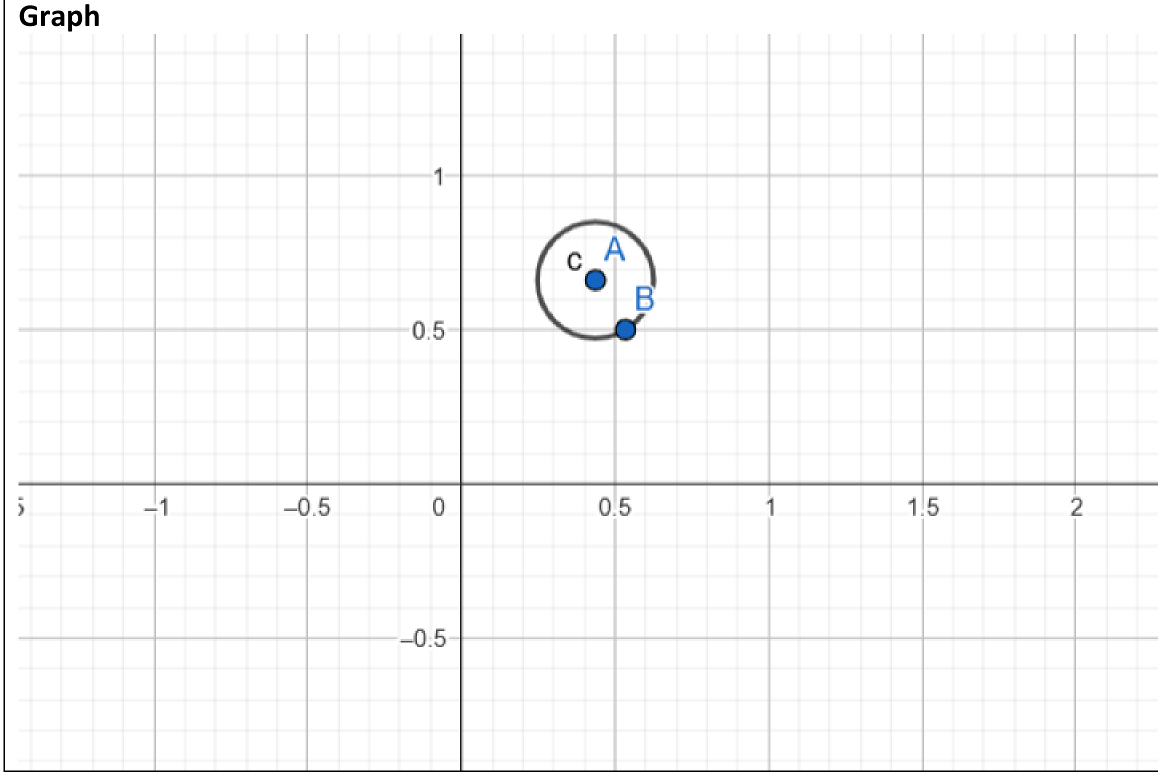




Question-

Create a simple day and night scenario using keyboard interaction. The key ‘D’ or ‘d’ will initiate the day mode and the key ‘N’ or ‘n’ will initiate the night mode.

Graph

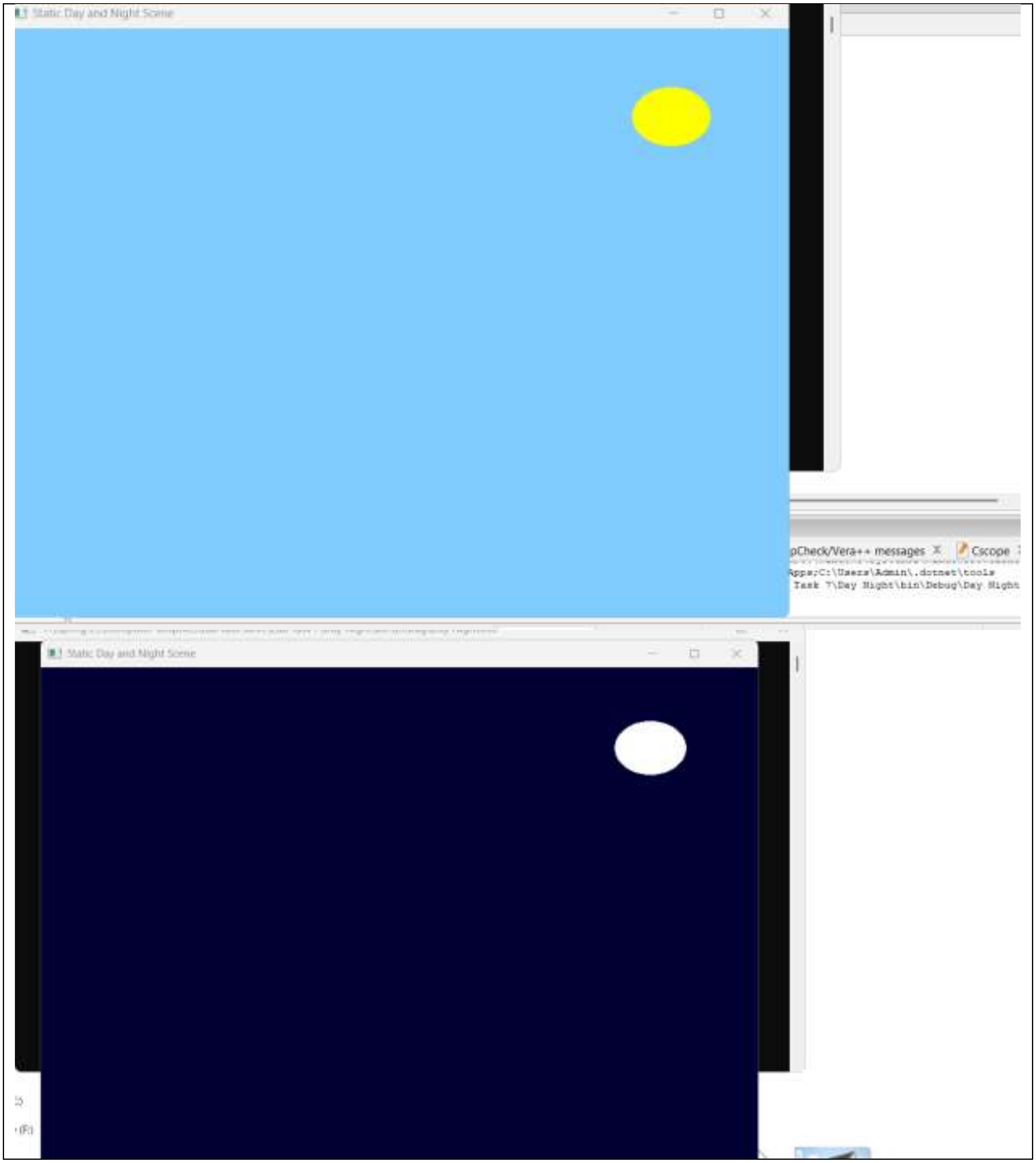


Code-

```
#include <GL/glut.h>
#include <cmath>
using namespace std;

bool isDay = true; // Start with day

void drawCircle(float x, float y, float radius) {
  int triangleAmount = 50;
  float twicePi = 2.0f * 3.1416f;

  glBegin(GL_TRIANGLE_FAN);
  for(int i = 0; i <= triangleAmount; ++i) {
    glVertex2f(
      x + (radius * cos(i * twicePi / triangleAmount)),
      y + (radius * sin(i * twicePi / triangleAmount))
    );
  }
  glEnd();
}

void display() {
  glClear(GL_COLOR_BUFFER_BIT);

  // Background
  if (isDay)
    glClearColor(0.5f, 0.8f, 1.0f, 1.0f); // Day
  else
    glClearColor(0.0f, 0.0f, 0.2f, 1.0f); // Night

  glClear(GL_COLOR_BUFFER_BIT);

  // Sun or Moon
  if (isDay)
    glColor3f(1.0f, 1.0f, 0.0f); // Sun (yellow)
  else
    glColor3f(1.0f, 1.0f, 1.0f); // Moon (white)

  drawCircle(0.7f, 0.7f, 0.1f); // Static top-right

  glutSwapBuffers();
}

void handleKeypress(unsigned char key, int x, int y) {
```

```
    if (key == 'd' || key == 'D') {
        isDay = true;
    } else if (key == 'n' || key == 'N') {
        isDay = false;
    }
    glutPostRedisplay(); // Redraw the scene
}

void init() {
    glMatrixMode(GL_PROJECTION);
    gluOrtho2D(-1, 1, -1, 1); // 2D view
}

int main(int argc, char** argv) {
    glutInit(&argc, argv);
    glutInitDisplayMode(GLUT_DOUBLE | GLUT_RGB);
    glutInitWindowSize(800, 600);
    glutCreateWindow("Day and Night Mode Switch");

    init();
    glutDisplayFunc(display);
    glutKeyboardFunc(handleKeypress); // Handle D/d/N/n keys
    glutMainLoop();

    return 0;
}
```

Output Screenshot (Full Screen)-


Static Day and Night Scene
pCheck/Vera++ messages
Cscope
Apps;C:\Users\Admin\.dotnet\tools
Task 7\Day Night\bin\Debug\Day Night
Static Day and Night Scene